

Machine Learning Algorithms

Systems Which Help Machine Learn

Units

4.1 Understanding Machine learning basics

4.2 Introduction to Common Supervised Learning Algorithms

4.3 Introduction to Common Unsupervised Learning Algorithms

4.4 Model Evaluation Metrics

OVERVIEW : MODULE

Machine Learning Algorithms

Learning Outcomes

- Understand the fundamental concepts of Machine Learning and its significance in various industries
- Differentiate between supervised, unsupervised, and reinforcement learning approaches
- Recognize key components of Machine Learning models, including features, labels, training data, and testing data
- Evaluate model performance using common metrics such as accuracy, precision, and recall.
- Implement basic Machine Learning algorithms (e.g., Linear Regression, Logistic Regression) using Python and relevant libraries



4.1

Understanding Machine Learning Basics

Lesson Learning Outcomes

- ✓ Understand the definition of a programming language and its necessity in AI & ML development
- ✓ Know what is Python and its history
- ✓ Understand why Python is important for AI & ML
- ✓ Know the names of important Python libraries

LESSON OVERVIEW

Module 5 Machine Learning Algorithms

Lesson 5.1 Understanding Machine Learning Basics

In this lesson, we will delve into the fundamental concepts of **Machine Learning (ML)**. We will begin by defining what ML is and exploring its significance across various industries. We will differentiate between the three main types of ML—supervised, unsupervised, and reinforcement learning

- Definition and significance of Machine Learning
- Types of Machine Learning: supervised, unsupervised, and reinforcement learning
- Explanation of features, labels, training data, and testing data

Understanding Machine Learning Basics

Introduction to Machine Learning

Machine learning is a field of Artificial Intelligence where computers learn from data and improve their performance over time without being explicitly programmed.

Importance of Machine Learning in Various Fields

Machine learning drives advancements in healthcare, finance, marketing, and more by automating tasks, making predictions, and uncovering patterns.

Examples of Real-World Applications

- **Recommendation Systems:** Think of personalized Netflix movie recommendations
- **Image Recognition:** Like facial recognition in smartphones or detecting objects in self-driving cars

Understanding Machine Learning Basics

Types of Machine Learning

Supervised Learning

In **Supervised learning**, the algorithm learns from a labeled dataset where each input example is associated with a corresponding output label. It aims to map input features to the correct output based on historical data.

Classification: Predicting discrete labels (e.g., spam vs. not spam).

Regression: Predicting continuous values (e.g., house prices).

Unsupervised Learning

Unsupervised learning deals with unlabeled data. The algorithm identifies patterns, structures, or relationships within the data without explicit output labels. Clustering and dimensionality reduction are common tasks

Clustering: Grouping similar data points (e.g., customer segmentation).

Dimensionality Reduction: Reducing the number of features while preserving information.

Understanding Machine Learning Basics

Types of Machine Learning

Reinforcement Learning

Reinforcement learning (RL) involves an agent interacting with an environment to learn optimal actions. The agent receives feedback (rewards or penalties) based on its actions and adjusts its strategy to maximize cumulative rewards.

Game Playing: Training agents to play games (e.g., AlphaGo, chess).

Robotics: Teaching robots to perform tasks through trial and error.

Understanding Machine Learning Basics

Basic Concepts in Machine Learning

Features and Labels

Features: Input variables (e.g., age, temperature).

Labels: Output or target variable to predict (e.g., price, category).

Training Data and Testing Data

Training Data: Used to train the model.

Testing Data: Independent data to evaluate model performance.

Model Evaluation Metrics

Metrics for regression: MSE, RMSE, MAE

Metrics for classification: Accuracy, Precision, Recall, F1 Score

Overfitting and Underfitting

Overfitting: Model too complex, fits noise.

Underfitting: Model too simple, lacks accuracy.

Cross-validation

Technique to estimate model performance on unseen data.

Helps prevent overfitting and assess generalization.

Understanding Machine Learning Basics

Introduction to Algorithms

Algorithms in the Context of Machine Learning

Machine learning algorithms are computational models that allow computers to understand patterns and make predictions based on data without explicit programming.

These algorithms process input data, learn from it, and generalize to make predictions or classifications.

Role of Algorithms in Training Models

- During training, the algorithm adjusts model parameters based on input data and target outcomes
- The goal is to minimize prediction errors and improve the model's ability to generalize to unseen data

Understanding Machine Learning Basics

SUMMARY

- ✓ Definition and significance of Machine Learning.
- ✓ Importance across various industries and domains.
- ✓ Supervised Learning, Unsupervised Learning, Reinforcement Learning.
- ✓ Explanation of each type with simple examples.
- ✓ Features and Labels.
- ✓ Training Data and Testing Data.
- ✓ Model Evaluation Metrics: accuracy, precision, recall.
- ✓ Overfitting and Underfitting, techniques like cross-validation.

4.2

Introduction to Common Supervised Learning Algorithms

Lesson Learning Outcomes

- ✓ Understand the basic concepts and applications of supervised learning
- ✓ Describe the working principles of Linear Regression, Logistic Regression, and Decision Trees
- ✓ Identify scenarios where each algorithm is suitable and effective
- ✓ Recognize the importance of supervised learning algorithms in machine learning applications

LESSON OVERVIEW

Module 5 Machine Learning Algorithms

Lesson 5.2 Introduction to Common Supervised Learning Algorithms

In this lesson, we will explore some of the most commonly used **supervised learning algorithms**. Supervised learning involves learning a mapping from input features to output labels based on labeled training data.

We will discuss three fundamental supervised learning algorithms: **Linear Regression**, **Logistic Regression**, and **Decision Trees**.

- Explore commonly used supervised learning algorithms
- Supervised learning entails mapping input features to output labels using labeled training data
- Discuss three fundamental supervised learning algorithms: Linear Regression, Logistic Regression, and Decision Trees

Introduction to Common Supervised Learning Algorithms

Linear Regression

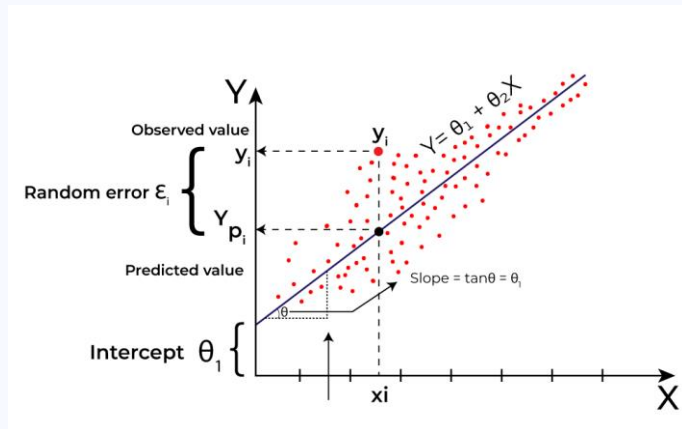
Linear regression is a linear approach to modeling the relationship between a dependent variable and one or more independent variables.

Formula:

$y = mx + b$ where y is the dependent variable, x is the independent variable, m is the slope, and b is the intercept.

Example:

Predicting house prices based on features like square footage, number of bedrooms, and location.



Introduction to Common Supervised Learning Algorithms

Logistic Regression

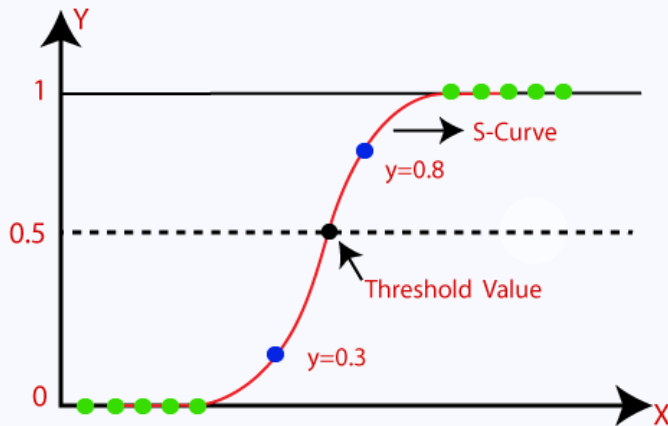
Logistic regression is a statistical method for analyzing a dataset in which there are one or more independent variables that determine an outcome.

Sigmoid Function:

Logistic regression uses the sigmoid function to map predictions to probabilities between 0 and 1.

Example:

Predicting the probability of a customer clicking on an ad based on features like age, gender, and browsing history.



Introduction to Common Supervised Learning Algorithms

Decision Trees

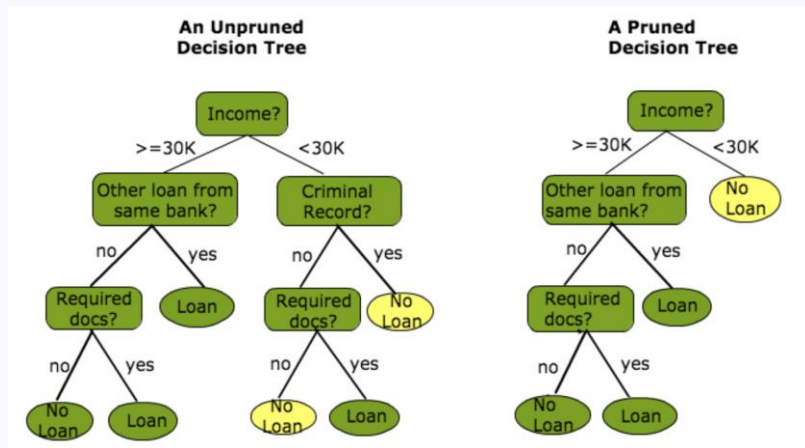
Decision trees are a popular supervised learning algorithm used for both classification and regression tasks.

Structure:

Decision trees consist of nodes that represent features, edges that represent decisions, and leaf nodes that represent outcomes.

Example:

Classifying whether an email is spam or not based on features like sender, subject, and keywords.



Introduction to Common Supervised Learning Algorithms

SUMMARY

- ✓ Explored fundamental unsupervised learning algorithms for discovering patterns and structures within unlabeled data.
- ✓ Covered three key algorithms: K-Means Clustering, Hierarchical Clustering, and Principal Component Analysis (PCA).
- ✓ K-Means Clustering partitions data into K clusters based on similarity, useful for tasks like customer segmentation and image compression.
- ✓ Principal Component Analysis (PCA) identifies patterns in high-dimensional data and expresses them in fewer dimensions, facilitating visualization and noise reduction.
- ✓ These algorithms are essential for data exploration, pattern recognition, and anomaly detection across various domains.
- ✓ Understanding these techniques equips learners with valuable tools for extracting insights from complex datasets and solving real-world problems.

4.3

Introduction to Common Unsupervised Learning Algorithms

Lesson Learning Outcomes

- ✓ Understanding of unsupervised learning principles and algorithms
- ✓ Ability to apply unsupervised learning techniques such as clustering and dimensionality reduction
- ✓ Recognition of scenarios where unsupervised learning is applicable and beneficial
- ✓ Proficiency in using Python libraries like scikit-learn to implement unsupervised learning algorithms

LESSON OVERVIEW

Module 5 Machine Learning Algorithms

Lesson 5.3 Introduction to Common Unsupervised Learning Algorithms

In this lesson, we explore fundamental **unsupervised learning algorithms** essential for discovering patterns and structures within unlabeled data.

The presentation covers three key algorithms: **K-Means Clustering**, **Hierarchical Clustering**, and **Principal Component Analysis (PCA)**.

- Introduction to Common Unsupervised Learning Algorithms
- Definition and basic functionality of unsupervised learning
- Use cases: Clustering, Dimensionality Reduction, Anomaly Detection
- Three fundamental algorithms covered: K-Means Clustering, Hierarchical Clustering, PCA

Introduction to Common Unsupervised Learning Algorithms

K-Means Clustering

K-Means Clustering is a popular algorithm used for clustering unlabeled data into K clusters. It iteratively assigns data points to clusters based on the similarity of their features to cluster centroids.

Example:

- Grouping customers based on purchasing behavior
- Clustering news articles based on their content

Introduction to Common Unsupervised Learning Algorithms

Hierarchical Clustering

Hierarchical Clustering builds a tree-like hierarchy of clusters by recursively merging or splitting them based on similarity. It does not require the number of clusters to be specified in advance.

Example:

- Taxonomy of species based on genetic similarities
- Organizing web documents into a hierarchical structure

Introduction to Common Unsupervised Learning Algorithms

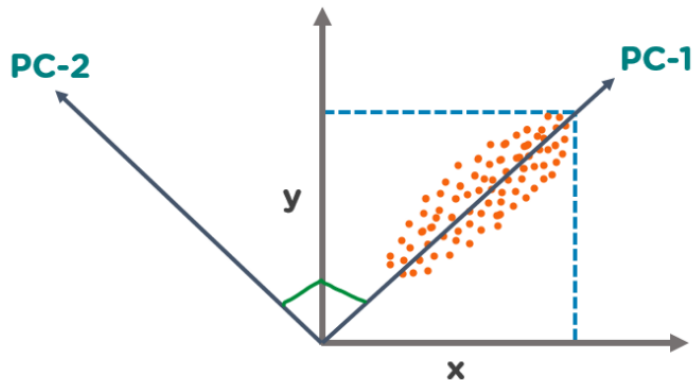
Principal Component Analysis (PCA)

Principal Component Analysis is a dimensionality reduction technique used to identify patterns in data and express them in fewer dimensions.

It finds the **orthogonal axes** (principal components) that capture the maximum variance in the data.

Example:

- Reducing the dimensionality of high-dimensional data like images or genomic data
- Visualizing high-dimensional data in a lower-dimensional space



Introduction to Common Unsupervised Learning Algorithms

SUMMARY

- ✓ Explores fundamental supervised learning techniques.
- ✓ Focuses on mapping input features to output labels using labeled training data.
- ✓ Linear Regression: Predicts continuous outcomes through linear relationships.
- ✓ Logistic Regression: Models binary outcomes using probabilities.
- ✓ Decision Trees: Versatile models for classification and regression tasks.

4.4

Model Evaluation Metrics

Lesson Learning Outcomes

- ✓ Understand the importance of evaluating machine learning models
- ✓ Identify the key differences between classification and regression evaluation metrics
- ✓ Grasp the meaning and application of common classification metrics like accuracy, precision, recall, and F1-score
- ✓ Learn how Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) are used to evaluate regression models
- ✓ Develop the ability to select appropriate evaluation metrics based on the specific task and business goals

LESSON OVERVIEW

Module 5 Machine Learning Algorithms

Lesson 5.4 Model Evaluation Metrics

The key to unlocking a model's power lies in understanding its **performance**.

This session explores essential metrics to evaluate your machine learning models, guiding you to identify **strengths**, **weaknesses**, and choose the **best model** for the job.

- Key metrics to assess how well your machine learning model performs
- Grasp the different metrics used for sorting data (classification) vs. predicting values (regression)
- Unpack accuracy, precision, recall, and F1-score to understand how well your model classifies data
- Explore MSE and RMSE to measure the error in predicting continuous values

Model Evaluation Metrics

Introduction To Evaluation Metrics

Machine learning models make predictions, but are those predictions accurate?

Evaluation metrics are essential tools for assessing model performance. By analyzing these metrics, we can identify a model's strengths and weaknesses, and compare different models

Model Evaluation Metrics

Classification vs. Regression Metrics

The type of metrics we use depends on the task our model is performing.

Classification models categorize **data points** (e.g., spam vs. not spam).

- Use metrics like accuracy, precision, recall, and F1-score.

Regression models predict **continuous values** (e.g., house prices).

- Use metrics like Mean Squared Error (MSE) and Root Mean Squared Error (RMSE).

Model Evaluation Metrics

Classification Metrics Explained

Accuracy: Overall correctness (correct predictions / total predictions)

- Formula:** $\text{Accuracy} = (\text{True Positive} + \text{True Negative}) / (\text{Total Population})$

Precision: How many positive predictions were actually correct? (true positives / all positive predictions)

- Formula:** $\text{Precision} = \text{True Positive} / (\text{True Positive} + \text{False Positive})$

Recall: How many actual positive cases did we identify? (true positives / all actual positive cases)

- Formula:** $\text{Recall} = \text{True Positive} / (\text{True Positive} + \text{False Negative})$

F1-Score: Harmonic mean of precision and recall, balancing both metrics

- Formula:** $\text{F1-Score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$

ACTUAL
If patient have cancer or not

		have cancer	doesn't have cancer
PREDICTION <i>what our model predicted</i>	have cancer	number of TP	number of FP
	doesn't have cancer	number of FN	number of TN

Model Evaluation Metrics

Regression Metrics Explained

Mean Squared Error (MSE)

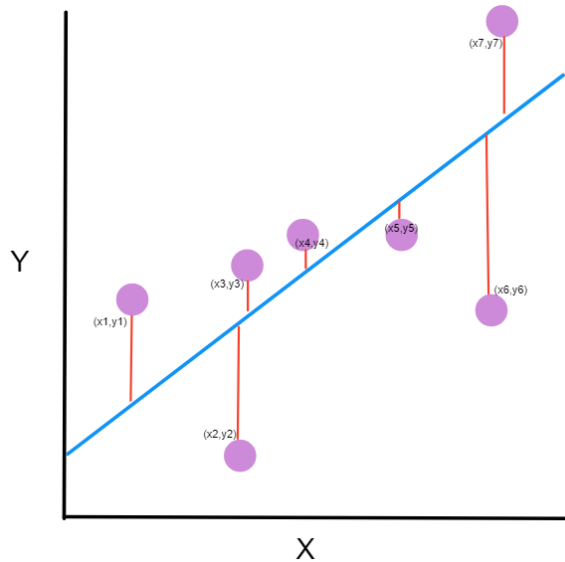
Average squared difference between predicted and actual values.

- Formula:** $MSE = (1/n) * \sum (y_i - \hat{y}_i)^2$ (where n is the number of data points, y_i is the actual value, and $\hat{y}_i$ is the predicted value)

Root Mean Squared Error (RMSE)

Square root of MSE (easier to interpret units).

- Formula:** $RMSE = \sqrt{MSE}$





4.5

Code Example

Implementing a Simple Linear Regression Model in Python

Model Evaluation Metrics

SUMMARY

- ✓ We learned that evaluating machine learning models is crucial to understand their effectiveness and identify areas for improvement.
- ✓ We explored the concept of evaluation metrics, the tools that help us assess a model's performance.
- ✓ We distinguished between metrics used for classification tasks (sorting data) and regression tasks (predicting continuous values).
- ✓ We unpacked key classification metrics like accuracy, precision, recall, and F1-score, gaining insights into how they assess a model's ability to classify data points.
- ✓ We're ready to delve deeper into these metrics and see how they are used in practice.

Key Takeaways

SUMMARY

- ✓ Understanding Machine Learning Fundamentals
- ✓ Exploration of Common Supervised Learning Algorithms
- ✓ Exploration of Common Unsupervised Learning Algorithms
- ✓ Introduction to Model Evaluation Metrics
- ✓ Considerations and Use Cases